SGT UNIVERSITY

FACULTY OF ALLIED HEALTH SCIENCES

Clinical Evaluation – Biochemistry Department

Name - Sakshi
course - Bmlt (3 rd sem)
Reg. no. - 250562063

Submitted to - Ms. kiran

1. Introduction to the Biochemistry Department

Clinical Biochemistry

Clinical biochemistry is a branch of laboratory medicine concerned with the chemical and biochemical examination of biological specimens, especially blood, serum, plasma, urine and other body fluids. It provides objective laboratory information that supports clinical diagnosis, assessment of disease, monitoring of treatment and evaluation of organ function. The department combines scientific principles with standardized laboratory procedures to produce accurate and clinically useful results.

Role of the Department

The clinical biochemistry department functions as an important diagnostic service in a hospital. Samples received from outpatient and inpatient areas are identified, checked for suitability, processed and analyzed using appropriate laboratory systems. Results are reviewed before reporting so that clinicians can use them along with the patient's history and other investigations.

Clinical Laboratory Workflow

The work of a biochemistry laboratory can be broadly divided into pre-analytical, analytical and post-analytical phases. The pre-analytical phase includes patient identification, sample collection, labelling, transport, reception and preparation. The analytical phase includes instrument operation, reagent handling, calibration, internal quality control and measurement. The post-analytical phase includes result verification, reporting, documentation and communication of critical or unusual results.

Importance of Standardization

Standard operating procedures, equipment maintenance, reagent storage, calibration and quality-control practices are necessary to minimize laboratory errors. Proper documentation and traceability also help maintain consistency between different runs and different days of laboratory work.

2. Clinical Biochemistry and Immunoassay

Clinical Biochemistry – Concept

Clinical biochemistry primarily uses chemical and physical measurement principles to determine the concentration or activity of substances in biological samples. Modern clinical chemistry analyzers automate several steps of the analytical process, including sample identification, reagent dispensing, reaction management, measurement and result calculation.

Immunoassay – Concept

Immunoassay is a laboratory analytical technique based on the specific interaction between an antigen and an antibody. Because antibodies can recognize particular molecular targets, immunoassays are used for the detection or measurement of substances that may be present in very small concentrations. The analytical signal is produced through an appropriate label and detection system.

Chemistry Analyzer and Immunoassay Analyzer

A clinical chemistry analyzer generally performs reactions in which a measurable physical or optical change is related to the concentration of an analyte. An immunoassay analyzer, in contrast, uses antigen–antibody binding as the central recognition step and converts this specific interaction into a measurable signal. Modern laboratory systems may combine automation, sample handling, reaction control, washing and signal detection within one platform.

Chemiluminescent Immunoassay

Chemiluminescent immunoassay uses a chemical reaction that produces light. The amount of emitted light is measured by a sensitive detector and is related to the analytical reaction. Such systems can provide high sensitivity and are suitable for analytes present at low concentrations.

General Immunoassay Principle

The basic sequence involves sample addition, interaction of the target substance with specific reagents, formation of an immune complex, removal or separation of unbound material where required, development of a detectable signal and measurement of that signal by the analyzer. Calibration converts the measured signal into a reportable result.

3. Laboratory Automation and Instrumentation

Introduction to Automated Analyzers

Automation in clinical laboratories reduces repetitive manual work and improves consistency of analytical operations. Automated systems can control sample volumes, reagent volumes, incubation conditions, reaction timing and measurement. They may also perform automatic checks and generate instrument flags when predefined conditions are not met.

Pre-analytical Considerations

Before a sample reaches an analyzer, correct patient identification and specimen selection are essential. Tubes should be appropriately labelled and inspected for problems such as clotting, insufficient volume, gross haemolysis or other unsuitable conditions. Serum or plasma may require centrifugation before analysis.

Analytical Phase

During analysis, the instrument performs a programmed sequence according to the assay or application selected. Important factors include reagent integrity, calibration, reaction temperature, timing, optical or other signal detection, and instrument cleanliness. Internal quality control is used to assess whether the analytical system is functioning within acceptable limits.

Post-analytical Phase

After measurement, results are checked for flags, analytical validity and consistency with quality-control status. Results may be transmitted electronically to the laboratory information system where available. Proper authorization and documentation are part of responsible result reporting.

Advantages of Automation

Major advantages include improved throughput, reduced manual pipetting, standardized reaction conditions, reduced transcription errors, better traceability and efficient handling of a large number of specimens. Automation does not remove the need for trained laboratory personnel; instead, it changes the focus toward monitoring, verification, troubleshooting and quality management.

4. Vitro 5600 – Principle and Working

Introduction

Vitro 5600 is an automated clinical chemistry analyzer used for routine laboratory chemistry work. It is designed to process multiple patient specimens with controlled reagent addition, reaction management and measurement. The analyzer supports standardized and automated handling of biochemical analytical procedures.

Basic Principle

The instrument is based mainly on photometric measurement of chemical reactions. In a photometric method, a reaction produces a measurable change in absorbance or optical intensity. The analyzer detects this change using an optical system and, after applying the relevant calibration relationship, converts the measurement into a reportable analytical value.

Major Functional Components

The system includes sample handling components, reagent handling and dispensing mechanisms, reaction areas, mixing and incubation arrangements, an optical measurement system, washing or cleaning arrangements, a control interface and software for data processing. The exact configuration depends on the instrument setup and laboratory application.

General Working Procedure

After instrument start-up, the operator checks system status, reagent availability, consumables, waste arrangements and quality-control requirements. Patient samples are identified and loaded according to the laboratory workflow. The analyzer aspirates the required sample, combines it with the appropriate reagent and allows the reaction to proceed under controlled conditions. The optical system measures the resulting signal. Software processes the signal using calibration information and produces the corresponding result or instrument flag.



5. Vitro 250 – Principle and Working

Introduction

Vitro 250 is an automated clinical chemistry analyzer used for standardized processing of laboratory specimens. Its automation is intended to improve consistency in sample and reagent handling and to support efficient laboratory workflow.

Principle

The analyzer primarily uses photometric or colorimetric measurement principles in which the analytical reaction generates a measurable optical change. Depending on the programmed analytical method, the instrument measures the reaction signal and uses calibration data to calculate the corresponding result.

Working Sequence

The general sequence begins with sample identification and instrument readiness checks. The sample is aspirated in a controlled volume and combined with the required reagent or reagents. Mixing and reaction conditions are controlled by the analyzer. After the appropriate reaction period, the optical signal is measured. The software evaluates the signal, applies the calibration relationship and displays the result or relevant flag.

Operator Responsibilities

The operator is responsible for correct sample loading, reagent verification, calibration when required, quality-control review, routine maintenance and responding to instrument alerts. Reagents should be handled according to manufacturer instructions, including storage conditions and onboard stability requirements.

Importance in Laboratory Practice

Automated chemistry analyzers improve workflow by reducing repetitive manual operations and allowing consistent timing of analytical reactions. Their usefulness depends on proper quality assurance, preventive maintenance and trained operation.

6. Bio-Rad D-10 – Principle and Working

Introduction

Bio-Rad D-10 is an automated analytical system based on high-performance liquid chromatography (HPLC). HPLC separates components of a sample according to their interactions with a stationary phase and a mobile phase. The separated fractions are detected and represented as peaks or related analytical signals.

HPLC Principle

In HPLC, a liquid mobile phase carries the sample through a chromatographic column. Different components interact differently with the column material and therefore move through the column at different rates. This produces separation. A detector records the separated components as they elute, allowing identification and measurement based on retention characteristics and calibration.

Working of the System

The sample is introduced into the analytical system and transported by the mobile phase through the chromatographic pathway. Components are separated in the column. The detector measures the signal generated by the separated fractions, and the software processes the chromatographic data. The final report is generated after the system applies the programmed analytical and calibration parameters.



7. MAGLUMI X3 – Immunoassay Analyzer

Introduction

MAGLUMI X3 is an automated chemiluminescence immunoassay system. It is designed around immunological recognition and sensitive light-based detection. In an immunoassay, the specific binding between an antigen and antibody provides the selectivity of the method.

Principle of Chemiluminescence

Chemiluminescence is the production of light during a chemical reaction. In an automated immunoassay system, a label or detection chemistry is associated with the immune reaction. After the required reaction and processing steps, the system produces a measurable light signal. The analyzer detects the signal using a sensitive optical detector.

General Working

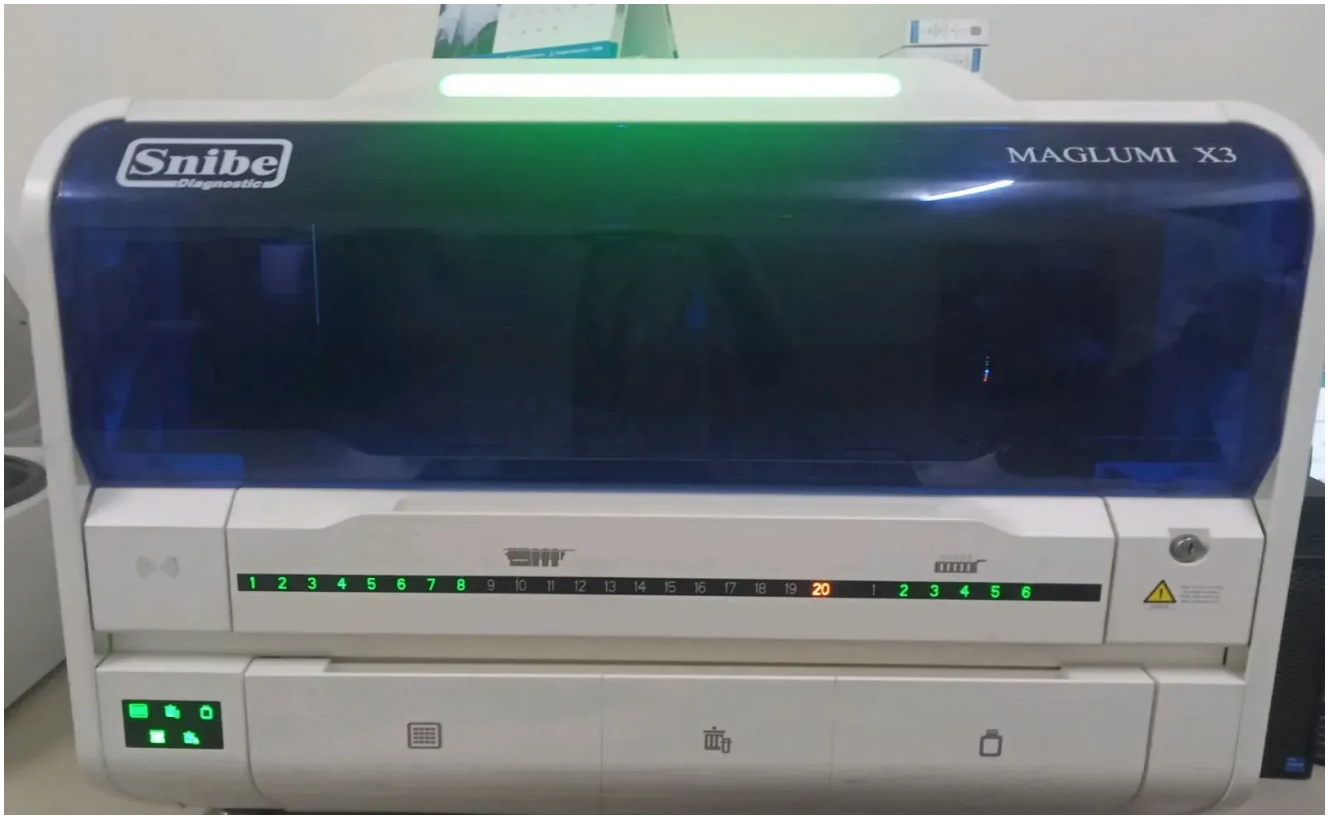
Patient samples and reagents are identified and processed by the automated system. The target substance interacts with specific immunological reagents. Depending on the assay design, the immune complex is formed and processed so that the appropriate labelled signal can be measured. The system performs the required reaction and separation steps, develops the chemiluminescent signal and measures the emitted light.

Signal and Result Calculation

The measured signal is interpreted using the calibration curve or analytical relationship established for the assay. The software converts the detected signal into the appropriate reportable result and may display flags when predefined analytical conditions are not met.

Advantages of Automated Immunoassay

Automation provides controlled reaction timing, standardized reagent handling, reduced manual intervention and efficient processing of specimens. Because immunoassays can detect low concentrations of target molecules, sensitive signal detection and careful quality control are particularly important.



8. TRACE 300 and Centrifuge

TRACE 300 – Introduction

TRACE 300 is used as an automated laboratory instrument within the clinical laboratory workflow. Automated systems of this type are designed to provide standardized processing and measurement according to the applications installed and validated by the laboratory.

General Working Concept

The instrument receives appropriately identified samples and processes them according to programmed analytical instructions. The system controls sample movement or aspiration, reagent or reaction steps where applicable, measurement and software-based result processing. The operator monitors instrument status, reagent or consumable requirements, quality-control information and system alerts.

Centrifuge – Introduction

A centrifuge is a laboratory instrument that separates components of a sample by applying centrifugal force. When a specimen is rotated at high speed, components with different densities move at different rates, allowing separation.

Principle of Centrifugation

Centrifugal force causes denser particles or cellular components to move away from the axis of rotation, while less dense components remain toward the upper portion of the tube. Correct balancing of tubes is essential for safe and effective centrifugation.

General Procedure

Appropriately labelled tubes are placed in the centrifuge with equal and balanced opposing loads. The required speed and time are selected according to the laboratory procedure. After centrifugation, the instrument is allowed to stop completely before the lid is opened. The separated serum or plasma is handled carefully to avoid disturbing the cellular layer.



9. Sample Processing, Quality and Safety

Sample Identification and Reception

Correct patient identification is the first requirement of reliable laboratory testing. On receipt, the sample should be checked against the request information, specimen type, volume, container, collection time where relevant and visible specimen quality. Any unsuitable specimen should be managed according to the laboratory's rejection or acceptance criteria.

Serum and Plasma Preparation

When serum is required, blood is allowed to clot appropriately before centrifugation. For plasma, blood is collected into the appropriate anticoagulant tube and processed according to the laboratory protocol. Separation should be performed carefully to reduce contamination between cellular and liquid components.

Internal Quality Control

Internal quality control uses control materials with known or expected characteristics to monitor analytical performance. Control results are reviewed before patient results are accepted according to the laboratory's quality-control policy. Trends, shifts and unexpected control values may indicate analytical problems requiring investigation.

Calibration and Maintenance

Calibration establishes the relationship between instrument response and the analytical value. Instruments also require routine cleaning, preventive maintenance, inspection and documentation. Maintenance schedules should follow the manufacturer's instructions and the laboratory's standard operating procedures.

Laboratory Safety

Biological specimens must be treated as potentially infectious. Appropriate personal protective equipment, hand hygiene, safe handling of sharps, proper waste segregation and disinfection procedures are required. Spills and exposure incidents should be managed according to institutional safety procedures.

10. Clinical Workload and Departmental Function

Monthly Patient Workload

The approximate monthly hospital workload provided for this clinical evaluation is 10,193 OPD patients and 8,293 IPD patients. Such a workload requires organized specimen reception, efficient sample processing, dependable instrumentation and proper communication between clinical areas and the laboratory.

Role of Biochemistry in Patient Care

Clinical biochemistry provides laboratory information that supports clinical decision-making. The value of a laboratory result depends not only on the analytical measurement but also on correct patient identification, appropriate specimen collection, reliable processing, quality control and accurate reporting.

Coordination with Clinical Areas

The laboratory receives specimens from different patient-care areas. Clear communication is required when a specimen is unsuitable, a repeat specimen is required or a result requires urgent attention according to institutional policy. Efficient coordination helps reduce delays in patient care.

Documentation

Laboratory documentation includes equipment records, maintenance records, calibration information, quality-control records, reagent information, sample processing records and reporting documentation. Proper records improve traceability and support laboratory quality management.

Professional Responsibilities

Medical laboratory professionals are responsible for following standard operating procedures, maintaining confidentiality, working safely, monitoring equipment performance and reporting results accurately. Technical competence and careful attention to detail are essential in a high-volume clinical laboratory.

11. Conclusion

Conclusion

The clinical evaluation of the Biochemistry Department provided an overall understanding of the organization and functioning of a modern clinical laboratory. Clinical biochemistry uses chemical and physical analytical principles to generate laboratory information, while immunoassay uses specific antigen–antibody interactions and sensitive detection systems.

Learning from Instrumentation

The evaluation covered the general principles and working concepts of the Vitro 5600, Vitro 250, Bio-Rad D-10, MAGLUMI X3 and TRACE 300, along with the role of the centrifuge in sample preparation. The instruments demonstrate how automation, photometric measurement, chromatography and chemiluminescent detection are integrated into contemporary laboratory practice.

Overall Clinical Significance

Reliable laboratory service depends on every stage of the workflow, from specimen identification and preparation to analytical measurement, quality control, result verification and reporting. Proper maintenance, calibration, documentation and safety practices are essential for maintaining the quality of laboratory services.